MIL-C-24643/29B
3 October 1986
SUPERSEDING
MIL-C-0024643/29A(SH)
8 November 1985
MIL-C-24643/29
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSMU (INCLUDING VARIATIONS LSMA AND LSMUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

Type LSMU cable can be manufactured in three variations: LSMU (unarmored), LSMA (armored) and LSMUS (double overall shield).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, in accordance with ASTM B 286, size 20-7, tin coated. Maximum diameter 0.039 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Ethylene propylene rubber or cross-linked polyethylene insulation, white. Minimum thickness of 0.011 inch. Standard identification code applied by method 1.
Fourth - The required number of singles (see table I) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward.
Fifth - Binder tape applied helically with overlap.

Construction variant for type LSMU

Sixth - Cross-linked polyolefin jacket. Thickness 0.030 inch minimum. Cable surface marking required.

(B) denotes changes.

AMSC N/A FSC 6145
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:05Z
Check the source to verify that this is the current version before use.

Construction variant for type LSMA

Sixth - Cross-linked polyolefin jacket. Thickness 0.030 inch minimum.
Seventh - Braided metal armor and paint.

Construction variant for type LSMUS

Sixth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.
Seventh - Spirally wrapped 0.002-inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.
Eighth - Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.
Ninth - Binder tape applied helically with overlap.
Tenth - Cross-linked polyolefin jacket. Thickness 0.030 inch minimum. Cable surface markings required.

TABLE I. Details.

Military 1/ part no. M24643/29	Size 2/	Number of conductors	Overall diameter (inches)					
			Type LSMU		Type LSMA		Type LSMUS	
			min	max	min	max	min	max
-01UN	14	14	0.365	0.400	0.415	0.450	0.425	0.460

1/ Unarmored. When armor option or double overall shield is required, see MIL-C-24643 for configuration of part number.
2/ Type LSMU, LSMA, and LSMUS.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	10.93
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Insulation resistance - megohms/1000 feet, minimum.....	500
Shield to shield (type LSMUS only), volts, minimum.....	6
Megohms/1000 feet, minimum.....	1.0
Jacket flaws.....	No failure
Conductor and shield (LSMUS only) continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:05Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Cold bending, cable - at 10 ± 2°C (6-inch maximum diameter mandrel)	No damage
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - (types LSMU and LSMUS) cycles, minimum	250
Armor (type LSMA only) - conformance to material, construction and coverage	No failure
Shield (type LSMUS only) - conformance to material, construction and coverage	No failure
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:05Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS: (Continued)

	Requirements
(B) Surface transfer impedance (type LSMUS only)	
Milliohms per meter, maximum	700
EMP response time - ns/500 feet	60

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH: 2500 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-29)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:05Z
Check the source to verify that this is the current version before use.